

MATHEMATICS SAMPLE QUESTIONS
for
BACHELOR OF SCIENCE DEGREE STUDENTS
[EXIT EXAM PREPARATIONS]

1. Which of the following vectors are **not** linearly independent?
 - A. $(0, 1, -1)$, $(1, 1, 0)$ and $(1, 0, 2)$
 - B. $(1, -1, 1)$, $(2, 0, -1)$ and $(2, -2, 2)$
 - C. $(2, 1, 0)$ and $(1, 1, 0)$
 - D. $(0, -1)$ and $(2, 1)$
2. Which one of the following type of matrix is not triangular?
 - A. Scalar matrix
 - B. Identity matrix
 - C. Zero/null matrix
 - D. Diagonal matrix
3. $\lim_{x \rightarrow 3} \frac{\sqrt{x^2 + 10} - \sqrt{19}}{x - 3}$ is equal to _____
 - A. $\frac{3}{\sqrt{19}}$
 - B. $\frac{6}{\sqrt{19}}$
 - C. 0
 - D. $\nexists$
4. Let $f(x) = x^2 + 2x + 1$. Then, $[f^{-1}(9)]'$ is:
 - A. $\frac{1}{\sqrt{6}}$
 - B. $\frac{1}{6}$
 - C. $\frac{1}{\sqrt{3}}$
 - D. $\frac{1}{3}$
5. The value of the indefinite integral $\int \cos^2 x \tan^3 x dx$ is:
 - A. $-\ln |\cos x| + \frac{1}{2} \cos^2 x + C$
 - B. $\ln |\sin x| + \frac{1}{2} \sin^2 x + C$
 - C. $-\ln |\cos x| + \frac{1}{2} \sin^2 x + C$
 - D. $\ln |\sin x| + \frac{1}{2} \cos^2 x + C$
6. If the interest rate per year is 16 and the compounding occurs every quarter, then the interest rate per compounding period is?
 - A. 0.04

- B. 0.004
C. 0.4
D. 40
7. Which of the following vector is perpendicular to both $A = (2, -3, 1)$ and $B = (1, 2, -4)$?
- A. $(4, 0, 1)$
B. $(10, 9, 7)$
C. $(3, 2, 0)$
D. $(0, 10, 9)$
8. The solution form of the ODE given by: $y'' + \frac{1}{x}y' - \frac{1}{x^2}y = 0$, for $x > 0$ is?
- A. x^r , r is constant
B. e^{xr} , r is constant
C. $x^{\frac{x}{r}}$, r is constant
D. $x^{\frac{1}{r}}$, r is constant
9. The assumption in calculating annuity is that every payment is
- A. Equal
B. Different
C. Nominal
D. Marginal
10. Let $f(x) = \sin(\cos x)$, then $f'(x)$ is equals to?
- A. $-\cos x \cos(\cos x)$
B. $-\sin x \cos(\cos x)$
C. $\sin x \cos(\cos x)$
D. $\cos x \cos(\cos x)$
11. What is the real rate of interest if the nominal rate is 10% and the inflation rate is 5%?
- A. 4.3%
B. 4.8%
C. 5.3%
D. 5.8%
12. The equation $(1 - x^2)y'' - 2xy' + 2(1 + x)y = 0$ is known as:
- A. Chebyshev equation
B. Bessel equation
C. Legendre equation
D. Bernoulli equation

13. Let a parabolic antenna takes the equation $x^2 = 160y$, then what is the area between $x^2 = 160y$, y -axis, $y = 2$ and $y = 4$?
- A. $\frac{3}{2}\sqrt{160}(8 + 2\sqrt{2})$
 B. $\frac{3}{2}\sqrt{160}(4 - 2\sqrt{2})$
 C. $\frac{3}{2}\sqrt{160}(8 - 2\sqrt{2})$
 D. $\frac{3}{2}\sqrt{160}(4 + 2\sqrt{2})$
14. A four year zero-coupon bond has 6% yields. What is its duration in years?
- A. 4
 B. 5
 C. 6
 D. 7
15. An investment of Birr 9,000 today will yield Birr 10,000 after one year. What is the Net Present Value if the interest rate is 10%?
- A. Birr 71
 B. Birr 81
 C. Birr 91
 D. Birr 101
16. $\lim_{x \rightarrow 0^+} [1 + \sin(4x))^{\cos(x)}]$ equals to:
- A. e^{-4}
 B. e^3
 C. e^{-3}
 D. e^4
17. Which of the following mapping T is NOT a linear transformation from $\mathbb{R}^3$ into $\mathbb{R}^2$?
- A. $T(x, y, z) = (x - y, x - z)$
 B. $T(x, y, z) = (z + x, y)$
 C. $T(x, y, z) = (1 + x, y)$
 D. $T(x, y, z) = (x + y + z, -z)$
18. Consider the following approximations in the table below,

Approximation	Error type
$f(x + h) \approx f(x) + hf'(x)$	A
$3.75697 \approx 3.757$	B
$3.75697 \approx 3.756$	C
$Height = 8 \pm 0.05cm$	D

what is the correct error type of A, B, C, D respectively?

- A. Truncation error, Chopping error, Round off error, and Absolute error.

- B. Truncation error, Round off error, Chopping error and Absolute error.
- C. Machine error, Chopping error, Round off error, and Absolute error.
- D. Machine error, Round off error, chopping error, and Absolute error.

19. Consider the fixed point iteration scheme.

$$x_{n+1} = g(x_n), \quad n = 0, 1, 2, \dots$$

what is the conditions for the convergence of such scheme to a fixed point $x = p$?

- A. $|g(x)| < 1$ for all x in a given interval
 - B. $|g(x)| > 1$ for all x in a given interval
 - C. $|g'(x)| > 1$, for all x in a given interval
 - D. $|g'(x)| < 1$, for all x in a given interval
20. Let A and B be 3×3 matrices with $\det(A) = 2$ and $\det(B) = -3$. Then which of the following is incorrect?
- A. $\det(2AB) = -48$
 - B. $\det(A^{-1}) = \frac{1}{2}$
 - C. $\det(A^t) = -2$
 - D. $\det(AB) = -6$
21. Which set G together with the operation indicated does not form a group.
- A. $G =$ set of all whole numbers, $a * b = a + b$.
 - B. $G =$ set of all integers, $a * b = a \times b$.
 - C. $G =$ set of rational numbers, $a * b = a + b$.
 - D. $G =$ set of real numbers $\neq -1$, $a * b = a - b$.
22. The value of $\iint (4xy - y^3) dA$ in a region bounded by $y = \sqrt{x}$ and $y = x^3$ is?
- A. $\frac{55}{156}$
 - B. -33
 - C. $\frac{5}{16}$
 - D. $-\frac{2}{15}$
23. Which of the following statement is false concerning any group G ?
- A. Its identity element is unique.
 - B. The inverse of every element of G is unique.
 - C. $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$.
 - D. $(a^{-1})^{-1} = a$ for all $a \in G$.
24. The value of $\int e^x [\sin(x) - \cos(x)] dx$ is?
- A. $-e^x \cos(x) + C$
 - B. $e^x \sin(x) + C$

- C. $-e^x \sec(x) + C$
 D. $-e^x \csc(x) + C$
25. Let $f(x) = \begin{cases} 2x^2 + b & \text{if } x \geq -1, \\ -x^3 & \text{if } x < -1. \end{cases}$, then what will be the value of b if $f(x)$ is continuous?
- A. 2
 B. -2
 C. 1
 D. -1
26. Which of the following statement is false?
- A. If G is a group with identity element e , then for any $a \in G$ there exists an $n \in \mathbb{Z}$ such that $a^n = e$.
 B. If G is an abelian group, then $(a * b)^n = a^n * b^n$ for all integers n .
 C. If G is a group in which $a^2 = e$ for all $a \in G$, then G is abelian.
 D. Any cyclic group is abelian.
27. If $\phi_1, \phi_2, \dots, \phi_n$ are n solutions of $y^{(n)} + a_1(x)y^{(n-1)} + \dots + a_1(x)y = 0$ on an interval I , then $\phi_1, \phi_2, \dots, \phi_n$ are linearly independent if and only if:
- A. $w(\phi_1, \phi_2, \dots, \phi_n) = 0, \forall x \in I$
 B. $w(\phi_1, \phi_2, \dots, \phi_n) \neq 0, \forall x \in I$
 C. $w(\phi_1, \phi_2, \dots, \phi_n) = 1, \forall x \in I$
 D. $w(\phi_1, \phi_2, \dots, \phi_n) = 2, \forall x \in I$
28. The maximum value of $f(x, y) = (xy)^6$ on the ellipse $\frac{x^2}{4} + y^2 = 1$ occurs at a point (x, y) for which y^2 is equal to
- A. $\frac{\sqrt{2}}{3}$
 B. $\frac{2}{3}$
 C. $\frac{5}{11}$
 D. $\frac{1}{2}$
29. If φ is a homomorphism from group G into group G' then, which of the following is incorrect?
- A. $\varphi(e) = e'$, the unit element of G' .
 B. $\varphi(a^{-1}) = \varphi(a)^{-1}$ for all $a \in G$.
 C. The image of φ is a subgroup of G .
 D. The kernel of φ is a subgroup of G .
30. Which one of the following is an integral domain?
- A. $\mathbb{Z}_6$

- B. $\mathbb{Z}_7$
- C. $\mathbb{Z}_8$
- D. $\mathbb{Z}_9$

31. Consider the following minimization problem

$$\begin{aligned} \text{Minimize : } Z &= 2x + 9y \\ \text{Subject to : } 3x + 5y &\geq 3 \\ 9x + 5y &\geq 8 \\ x \geq 0, y &\geq 0 \end{aligned}$$

What is the objective function of the dual maximization problem? Use u and w for the dual variables.

- A. $C = 9u + 3w$
 - B. $C = 3u + 8w$
 - C. $C = 9u + 3w$
 - D. $C = 5u + 5w$
32. If $\mathbf{x}$ is a decision variable of LPP and un restricted in sign, then this variable can be converted in to $\mathbf{x} = \mathbf{x}' - \mathbf{x}''$ so as to solve the LPP by simplex method, where:
- A. $x' \geq 0$ and $x'' \geq 0$
 - B. $x' \geq 0$ and $x'' \leq 0$
 - C. $x' \leq 0$ and $x'' \geq 0$
 - D. $x' \leq 0$ and $x'' \leq 0$
33. Before formulating a formula for LP model, it is better to
- A. Express each constraint in words.
 - B. Verbally identify decision variable
 - C. Express the objective function in words.
 - D. All of the above
34. In the simplex method, if in pivot column all the entries are negative or zero when choosing leaving variable, then
- A. Solution is degenerate
 - B. Alternative optimal solution
 - C. Unbounded solution
 - D. Solution is infeasible
35. While solving an LPP (defined by n variables and m equations, $m < n$) through simplex method, basic solutions are determined by setting $n - m$ variables equal to zero and solving m equations to obtain solution for remaining m variables, provided the resulting solutions are unique. This means that the maximum number of basic solutions is

- A. $\frac{n!}{(m-n)!m!}$
- B. $\frac{n!}{(m-n)!n!}$
- C. $\frac{n!}{(m+n)!m!}$
- D. $\frac{n!}{(m+n)!n!}$

36. The following expression for $f(x)$ describes

$$f(x) = \sum_{i=0}^n y_i \prod_{\substack{j=0 \\ j \neq i}}^n \left(\frac{x - x_i}{x_j - x_i} \right)$$

- A. Lagrange polynomial interpolation
- B. Cubic spline.
- C. Piecewise linear interpolation.
- D. General Newton's interpolation

37. Given a positive integer n , then (n) denotes:

- A. The sum of positive divisors
- B. The number of positive divisor of n
- C. The number of coprime numbers of n
- D. None

38. What is the general solution of the ODE $x^2y'' + 2xy' - 6y = 0$ ia?

- A. $c_1x + c_2x^2$
- B. $c_1x^{-3} + c_2x^2$
- C. $c_1x^2 + c_2x^{-4}$
- D. $c_1x^{-2} + c_2x^3$

39. $\sigma(10) =$ _____

- A. 5
- B. 9
- C. 10
- D. 18

40. The complete solution of the linear congruence $9x = 6(mod 15)$ is

- A. $\{4, 5, 14\}$
- B. $\{4, 9, 14\}$
- C. $\{3, 9, 14\}$
- D. $\{7, 6, 14\}$

41. What is the radius of convergence of the power series $\sum_1^\infty \left(\frac{n}{n+3} \right) \left(\frac{x+2}{2^n} \right)^n$?

- A. -4
 B. -2
 C. 2
 D. 4
42. For what values of p does the series $\sum_1^\infty \left(\frac{1}{\sqrt[3]{n^p + 1}} \right)$ converges?
 A. $p < 3$
 B. $p > \frac{2}{3}$
 C. $p < \frac{1}{3}$
 D. $p > 3$
43. The octal expansion of $(12345)_{10}$ is
 A. $(321481)_8$
 B. $(30071)_8$
 C. $(23415)_8$
 D. $(21567)_8$
44. The integral solution of the liner Diophantine equation $75x + 12y = 24$ is
 A. $\{8 + 4t, -6 - 25t\}$
 B. $\{8 + 4t, -48 - 25t\}$
 C. $\{8 - 4t, -48 - 25t\}$
 D. $\{8 - 4t, 48 + 25t\}$
45. If ϕ_p is the particular solution of $Ly = y^{(n)} + a_1(x)y^{(n-1)} + \dots + a_1(x)y = b(x)$, and $\phi_1, \phi_2, \dots, \phi_n$ are linearly independent solution of $Ly = 0$, then every solution ϕ of $Ly = b(x)$ is often takes the form:
 A. $\phi = \phi_p + c_1\phi_1 + c_2\phi_2 + \dots + c_n\phi_n$
 B. $\phi = \phi_p + c_1\phi_1 + c_2\phi_2^2 + \dots + c_n\phi_n^{(n-1)}$
 C. $\phi = \phi_p + c_1e^{\phi_1} + \dots + c_ne^{\phi_n^{(n-1)}}$
 D. $\phi = \phi_p + c_1\log(\phi_1) + \dots + c_n\log(\phi_n^{(n-1)})$
46. Let $f(x, y) = xy$ with $x = \cos t$ and $y = \sin t$, then which one of the following is the value of $\frac{df}{dt}$ at $t = \frac{\pi}{2}$?
 A. 1
 B. -1
 C. 0
 D. $\sqrt{2}$
47. Which one is the directional derivative of $f(x, y, z) = 4x - e^{3xz}$ in the direction of $\vec{v} = -i + 4j + 2k$ at $(3, -1, 0)$?

- A. $-\frac{14}{\sqrt{21}}$
- B. $-\frac{2}{\sqrt{10}}$
- C. $-\frac{21}{10}$
- D. $-\frac{14}{5}$

48. The curvature of a circle with radius a is

- A. a^2
- B. $\sqrt{a}$
- C. $\frac{1}{a}$
- D. $2a$

49. Which of the following is False about Jacobi and Gauss-Seidel iterative methods

- A. In Jacobi method the value of the variables is not modified until next iteration, whereas in Gauss-Seidel method the value of the variables is modified as soon as new value is evaluated.
- B. Jacobi method requires less number of iterations than Gauss-Seidel method to converge to the actual solution with a certain degree of accuracy.
- C. Jacobi or Gauss-Seidel method is an iterative method for solving equations of diagonally dominant system of linear equations.
- D. Sufficient condition for the convergence of the approximations obtained by both methods is that the system of equations is diagonally dominant.
- E. If the given system of equations is diagonally dominant, then any initial approximation will converge to the exact solution.

50. Which of the following methods usually gives smaller error in finding numerical derivatives?

- A. Forward difference method
- B. Backward difference method
- C. Central difference method
- D. They all give the same error